



Multilingual RAG

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Speaker



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01

RAG Review

RAG

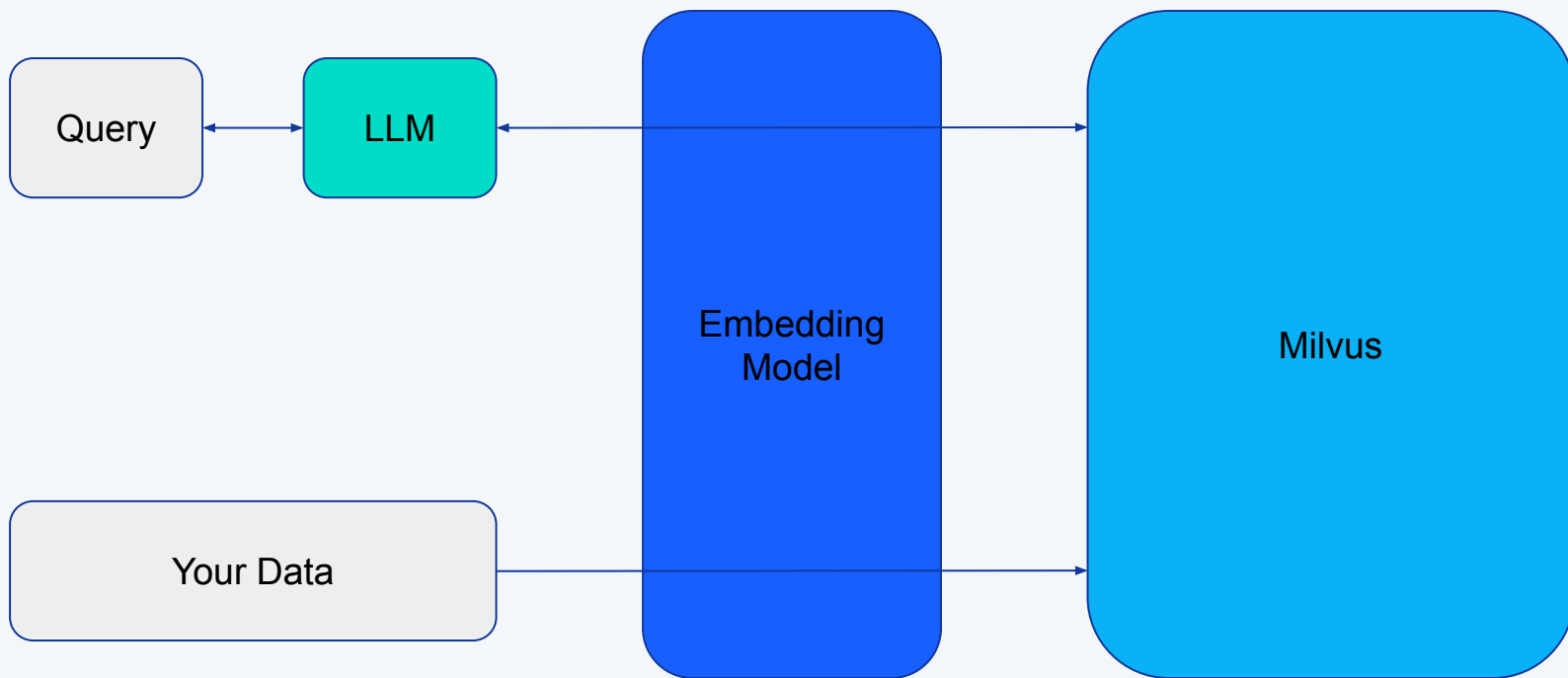
RAG

Inject your data via a vector

database like Milvus/Zilliz

Primary Use Case

- Factual Recall
- Forced Data Injection
- Cost Optimization

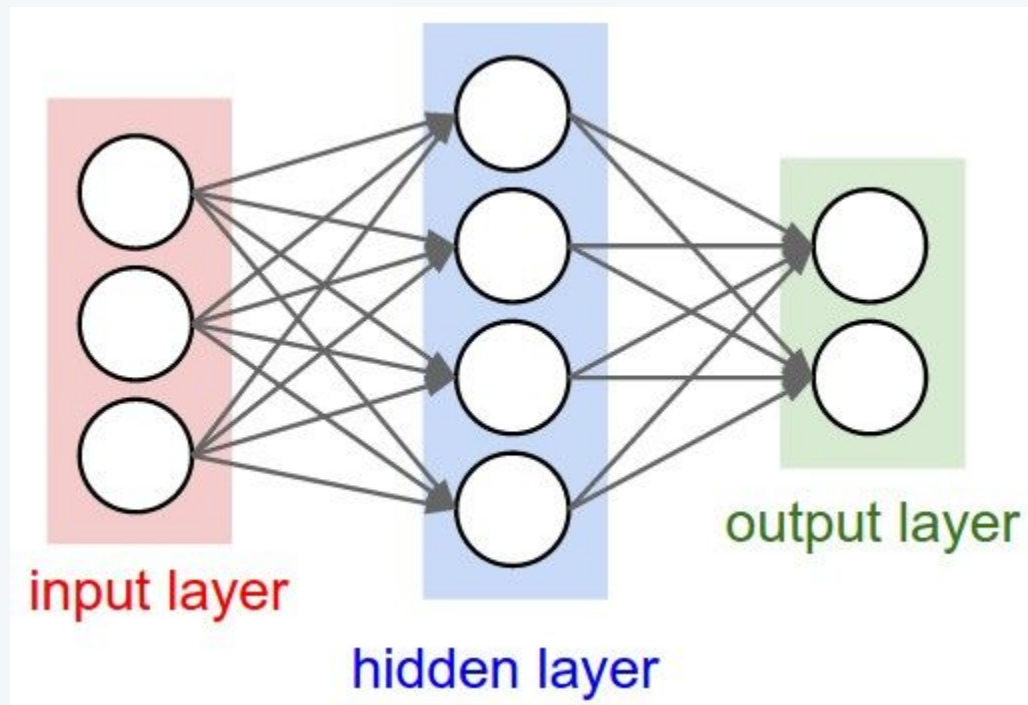


02

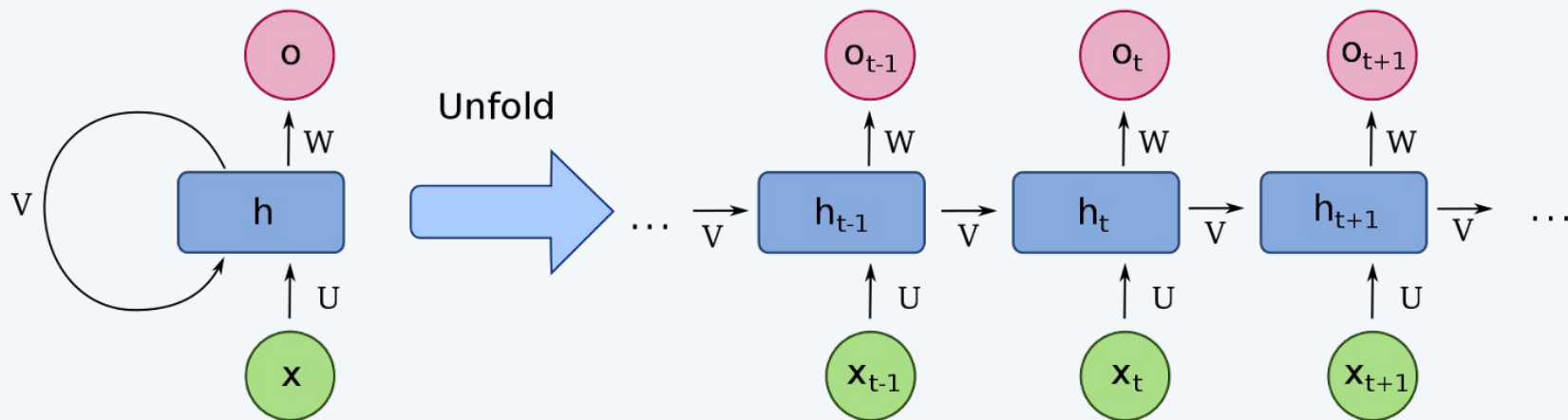
LLMs and Embedding Models

How did LLMs come about?

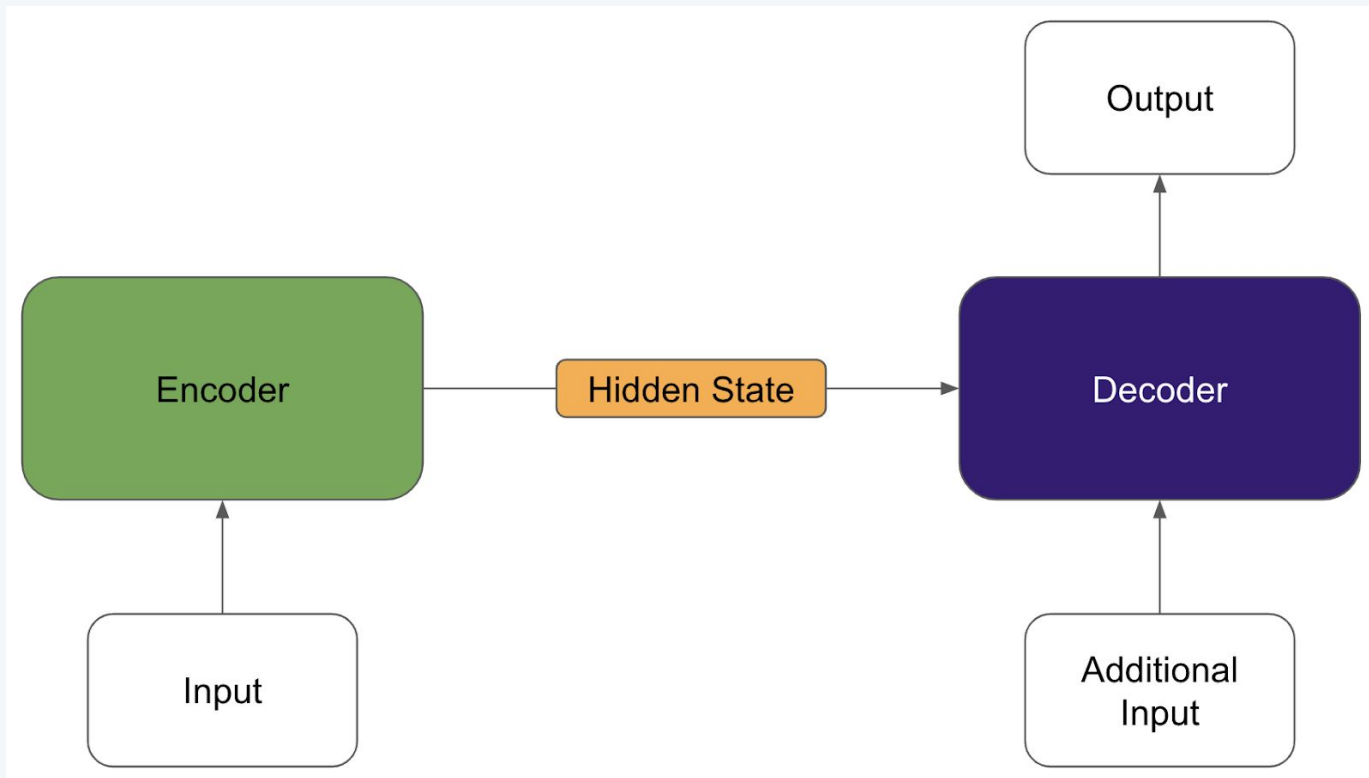
A Basic Neural Net



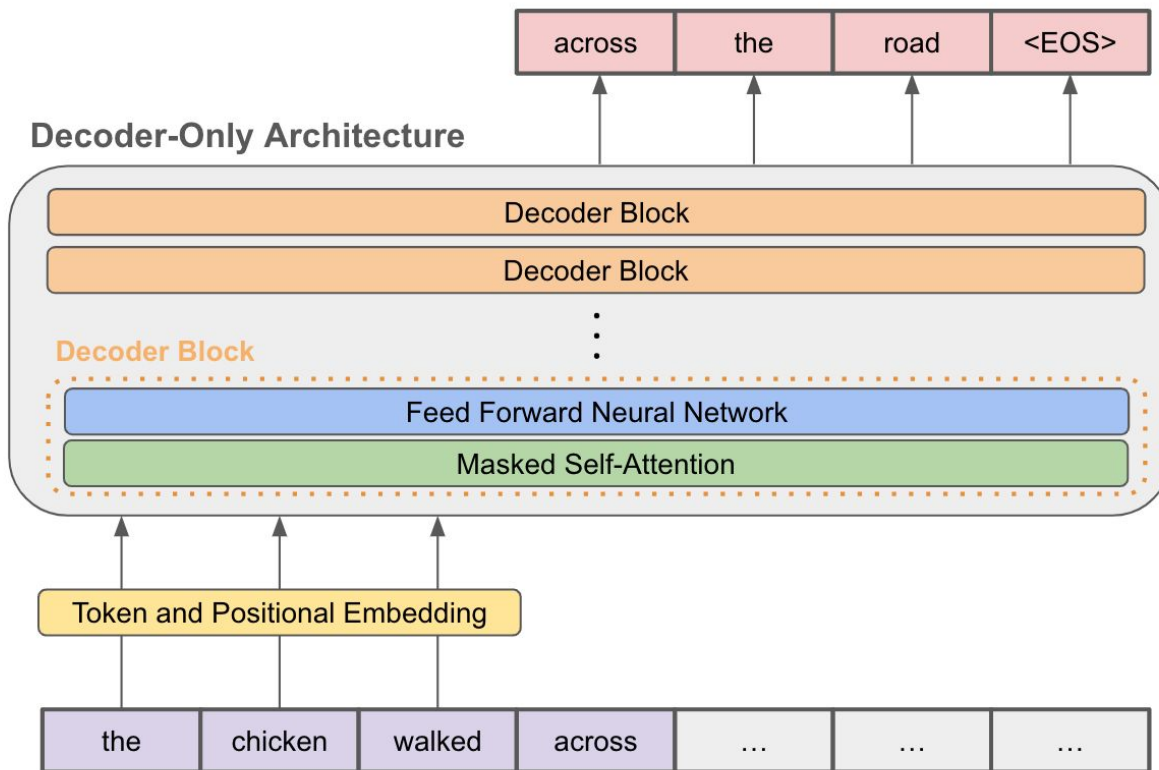
A Recurrent Neural Network



A Transformer Architecture

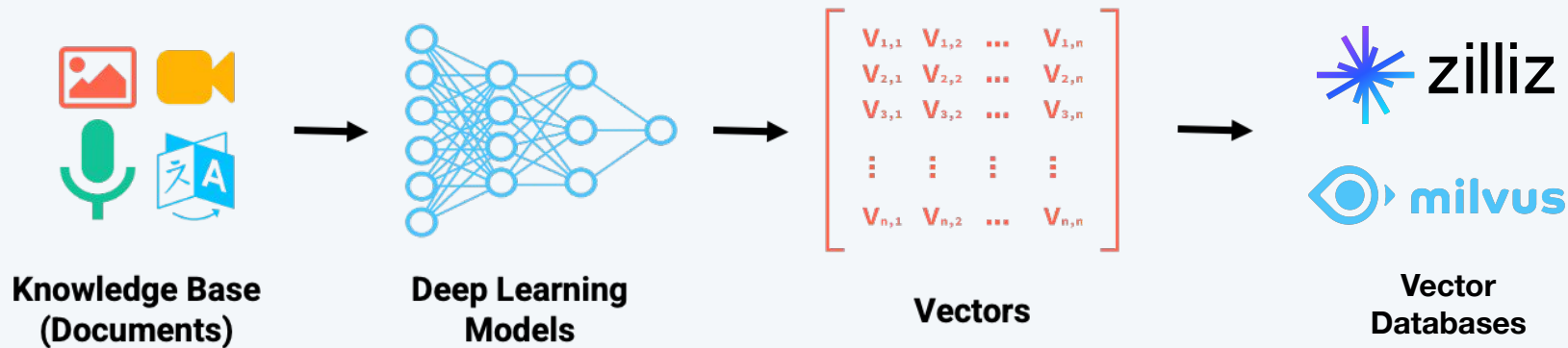


GPT



What about Embedding Models?

Deep Learning Models w/o Last Layer



LLMs

- Large models
- Generate text
- Reasoning capability
- Based on
transformers

Embedding Models

- Smaller
- Non predictive
- Non generative

03

Vector Databases

Find Semantically Similar Data

Apple made profits of \$97 Billion in 2023

I like to eat apple pie for profit in 2023

Apple's bottom line increased by record numbers in 2023

But wait! There's more!



Semantic Similarity

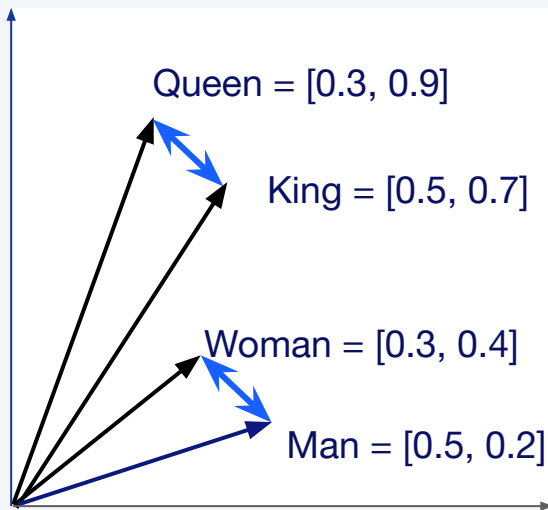
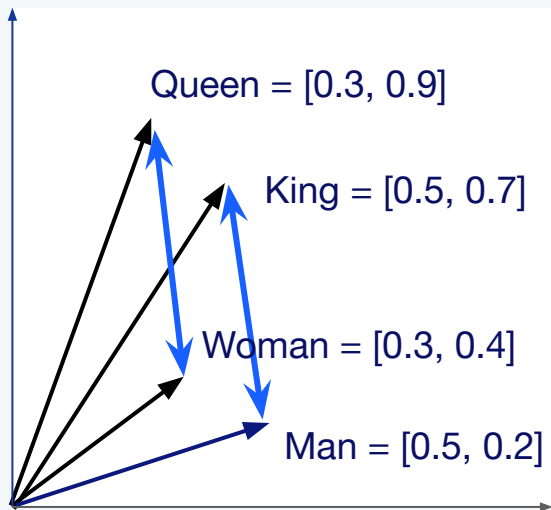


Image from [Sutor et al](#)

$$\text{Queen} - \text{Woman} + \text{Man} = \text{King}$$

$$\begin{array}{r} \text{Queen} = [0.3, 0.9] \\ - \text{Woman} = [0.3, 0.4] \\ \hline \end{array}$$

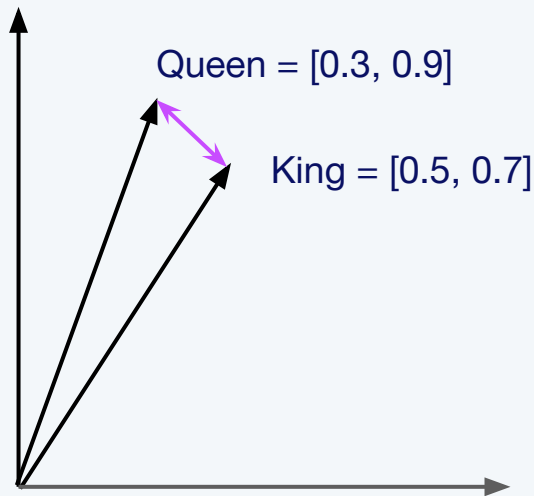
$$\begin{array}{r} [0.0, 0.5] \\ + \text{Man} = [0.5, 0.2] \\ \hline \end{array}$$

$$\text{King} = [0.5, 0.7]$$

**Similarity metrics are ways to measure distance in
vector space**

Vector Similarity Metric: L2 (Euclidean)

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

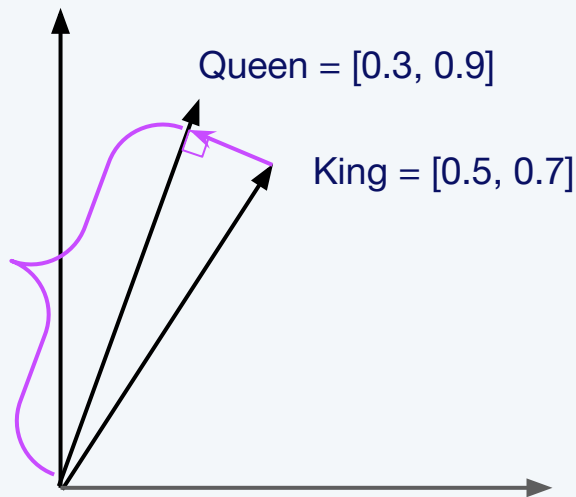


$$\begin{aligned} d(\text{Queen}, \text{King}) &= \sqrt{(0.3-0.5)^2 + (0.9-0.7)^2} \\ &= \sqrt{(0.2)^2 + (0.2)^2} \\ &= \sqrt{0.04 + 0.04} \\ &= \sqrt{0.08} \approx 0.28 \end{aligned}$$

Vector Similarity Metric: Inner Product (IP)

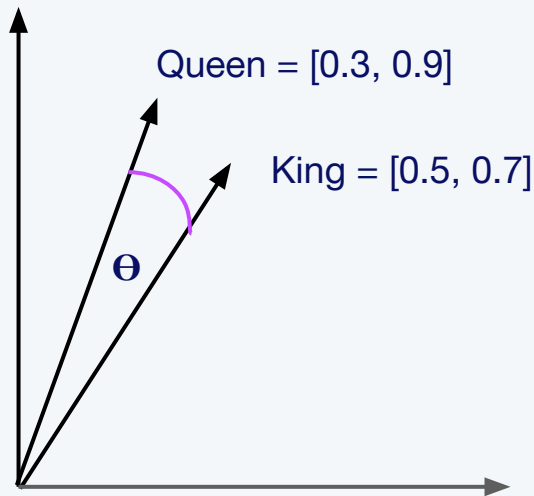
$$a \cdot b = \sum_{i=1}^n a_i b_i$$

$$\begin{aligned}\text{Queen} \cdot \text{King} &= (0.3 \cdot 0.5) + (0.9 \cdot 0.7) \\ &= 0.15 + 0.63 = 0.78\end{aligned}$$



Vector Similarity Metric: Cosine

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$



$$\cos(\text{Queen, King}) = \frac{(0.3 \cdot 0.5) + (0.9 \cdot 0.7)}{\sqrt{0.3^2 + 0.9^2} * \sqrt{0.5^2 + 0.7^2}}$$

$$= \frac{0.15 + 0.63}{\sqrt{0.9} * \sqrt{0.74}}$$

$$= \frac{0.78}{\sqrt{0.666}}$$

$$\approx 0.03$$

Vector Similarity Metrics

Euclidean - Spatial distance

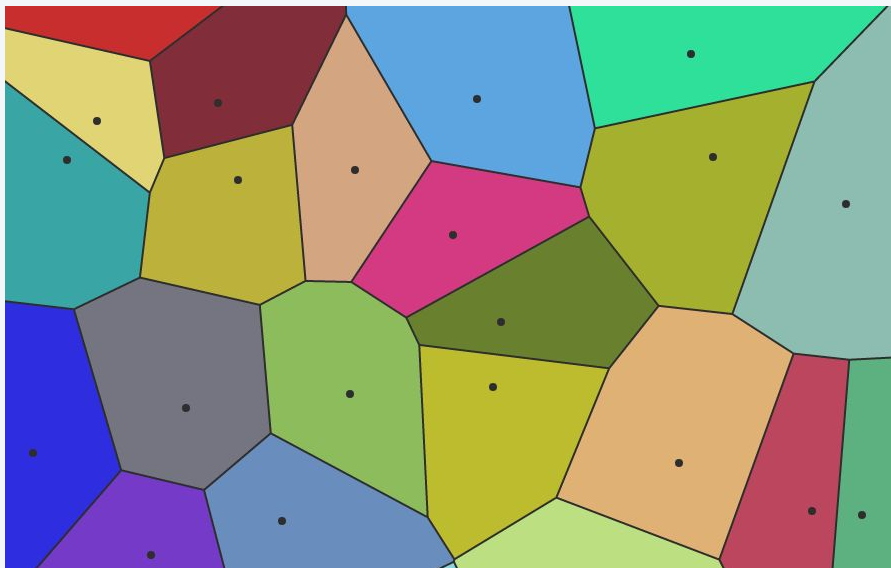
Cosine - Orientational distance

Inner Product - Both

With normalized vectors, $IP = \text{Cosine}$

Indexes organize the way we access our data

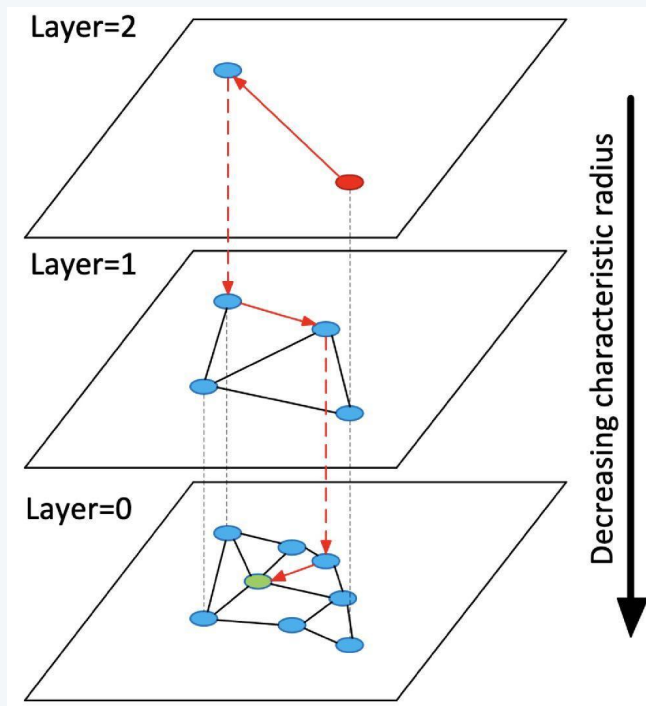
Inverted File Index



Source:

<https://towardsdatascience.com/similarity-search-with-ivfpq-9c6348fd4db3>

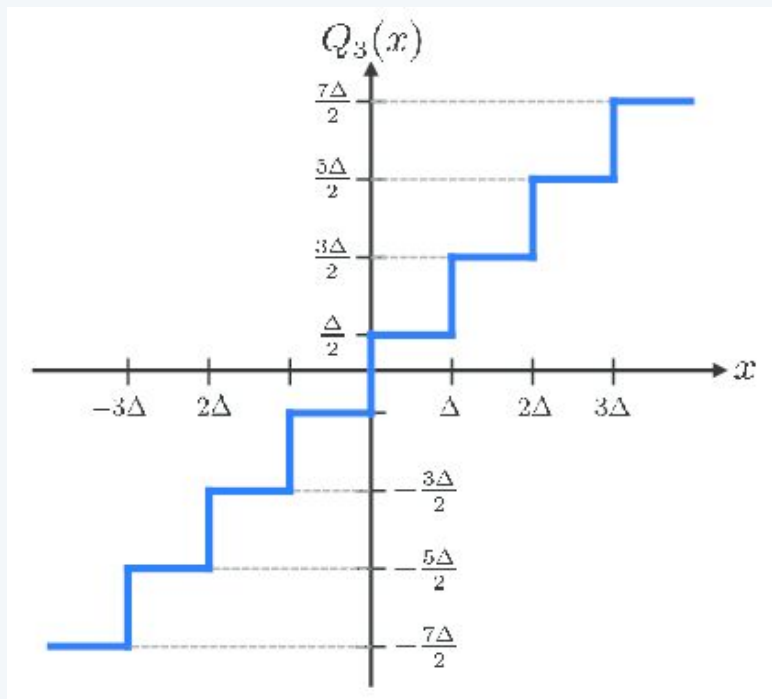
Hierarchical Navigable Small Worlds (HNSW)



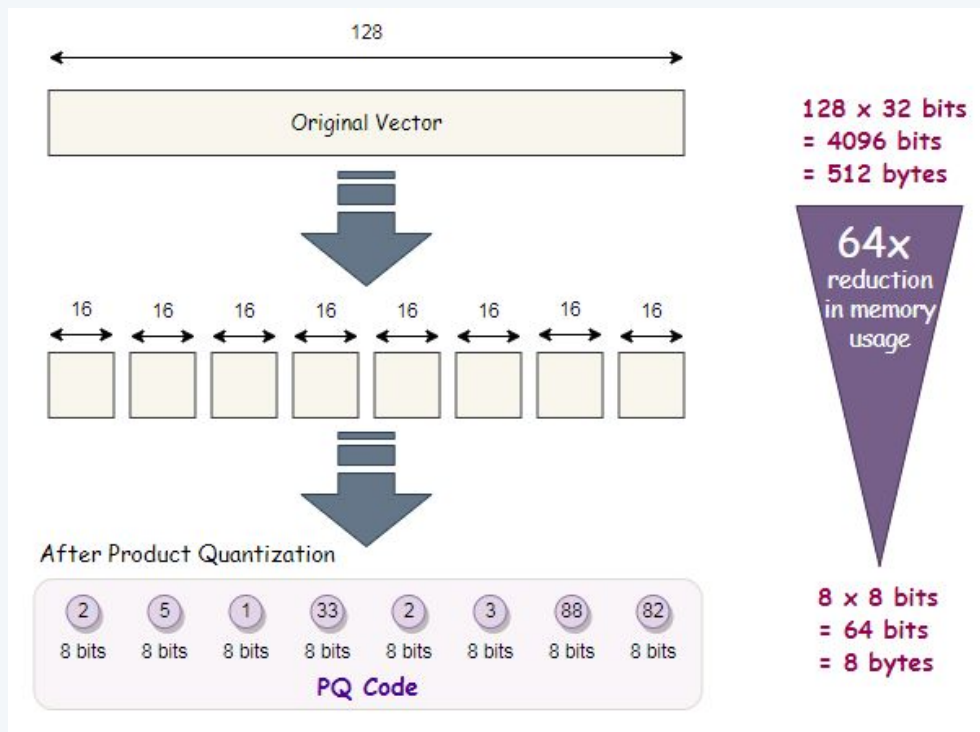
Source:

<https://arxiv.org/ftp/arxiv/papers/1603/1603.09320.pdf>

Scalar Quantization (SQ)



Product Quantization



Source:

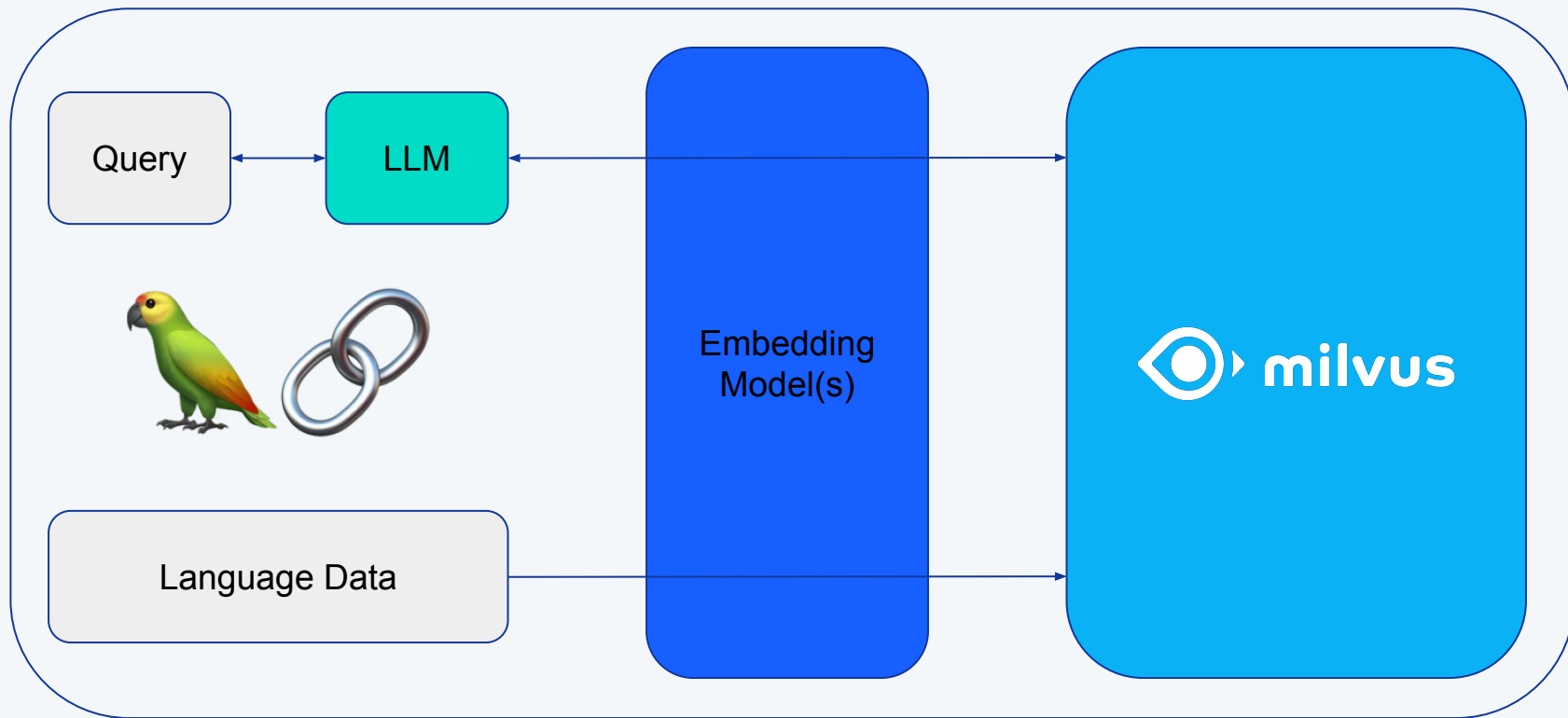
<https://towardsdatascience.com/product-quantization-for-similarity-search-2f1f67c5fddd>

Indexes Overview

- IVF = Intuitive, medium memory, performant
- HNSW = Graph based, high memory, highly performant
- Flat = brute force
- SQ = bucketize across one dimension, accuracy x memory tradeoff
- PQ = bucketize across two dimensions, more accuracy x memory tradeoff

04

Demo



RAG



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